

LECTURE SHEET

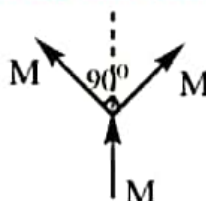
EXERCISE-I

(Bar magnet, Magnetic Moment)

Straight Objective Type Questions

1. Find the resultant magnetic moment for the following arrangement

- 1) $\sqrt{2}M$
 2) $(\sqrt{2} + 1)M$
 3) $(\sqrt{2} - 1)M$
 4) M



2. A bar magnet of magnetic moment '
- M_1
- ' is axially cut into two equal parts. If these two parts are arranged perpendicular to each other, the resultant magnetic moment is
- M_2
- . Then the value of
- M_1/M_2
- is

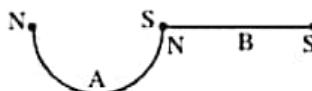
- 1) $\frac{1}{2\sqrt{2}}$ 2) 1 3) $\frac{1}{\sqrt{2}}$ 4) $\sqrt{2}$

3. In an iron bar
- $(5 \times 1 \times 1)\text{cm}^3$
- magnetic moment of an atom is
- $1.8 \times 10^{-23} \text{ Am}^2$
- . What is magnetic moment of the bar in the state of saturation. (Density =
- $7.8 \times 10^3 \text{ kg/m}^3$
- , atomic mass = 56 gm)

- 1) 6.54 Am^2 2) 7.54 Am^2 3) 4.54 Am^2 4) 3.54 Am^2

Numerical Value Type Questions

4. A magnet of length
- L
- and moment
- M
- is cut into two halves (A and B) perpendicular to its axis. One piece A is bent into a semicircle of radius
- R
- and is joined to the other piece at the poles as shown in the figure below.



Assuming that the magnet is in the form of a thin wire initially, the moment of the resulting magnet is given by $\frac{M(x + \pi)}{x\pi}$ then 'x' is

EXERCISE-II

(Magnetic induction, Coulomb law and Null points)

Straight Objective Type Questions

1. The length of a magnet of moment
- 5Am^2
- is 14cm. The magnetic induction at a point, equidistant from both the poles is
- $3.2 \times 10^{-5} \text{ wb/m}^2$
- . The distance of the point from each pole is

- 1) 5cm 2) 10cm 3) 25cm 4) 15cm

2. A magnet of length 10cm and magnetic moment
- 1Am^2
- is placed along the side AB of an equilateral triangle ABC. If the length of the side AB is 10cm, the magnetic induction at the point C is

- 1) 10^{-9}T 2) 10^{-7}T 3) 10^{-5}T 4) 10^{-4}T

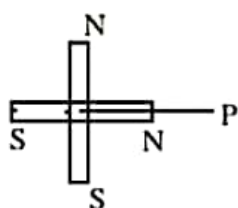
3. Two short bar magnet of magnetic moment
- M
- each are arranged at the opposite corners of a square of side 'd' such that their centres coincide with the corners and their axes are parallel. If the like poles are in same direction the magnetic induction at any of the other corner of the square is

- 1) $\frac{\mu_0 M}{4\pi d^3}$ 2) $\frac{\mu_0 2M}{4\pi d^3}$ 3) zero 4) $\frac{\mu_0 3M}{4\pi d^3}$

4. The magnetic induction at a point on the axial line of a short magnet is 10 Tesla east. The magnetic induction at the same point when the magnet is rotated in the anticlock wise direction by 90° and 180° are
 - 1) 10T south and 5T west
 - 2) 5T south and 10T west
 - 3) 20T south and 10T west
 - 4) 10T south and 20T west
5. A short bar magnet of magnetic moment $5.25 \times 10^{-2} \text{ JT}^{-1}$ is placed with its axis perpendicular to the earth's field direction. At what distance from the centre of the magnet, the resultant field is inclined at 45° with earth's field on (a) its normal bisector and (b) its axis. Magnitude of the earth's field at the place is given to be 0.45G. Ignore the length of the magnet in comparison to the distances involved.
 - 1) 2.8 cm, 5 cm
 - 2) 2 cm, 5 cm
 - 3) 8 cm, 1.2 cm
 - 4) 5 cm, 6.3 cm
6. Two short bar magnets P and Q are arranged such that their centres are on the X-axis and are separated by a large distance. The magnetic axes of P and Q are along X, Y-axis respectively. R lies midway between their centres. If B is the magnitude of induction due to Q at R, then magnitude of total induction at R due to both the magnets is
 - 1) 3B
 - 2) $\sqrt{5}B$
 - 3) $\frac{\sqrt{5}}{2}B$
 - 4) B
7. A short bar magnet is kept along the magnetic meridian with its north pole pointing north. A neutral point is located at distance 'd' from the centre. The ratio of magnetic inductions at a distance d/2 and 2d from the centre of the magnet on the equatorial line is
 - 1) 1 : 4
 - 2) 4 : 1
 - 3) 8 : 1
 - 4) 1 : 8
8. A bar magnet of length 8 cm and having a pole strength of 1.0 Am is placed vertically on a horizontal table such that with its south pole on table. A neutral point is found on table at a distance of 6 cm north of the magnet. Calculate B_H value.
 - 1) $11 \mu\text{T}$
 - 2) $22 \mu\text{T}$
 - 3) $33 \mu\text{T}$
 - 4) $33 \mu\text{T}$
9. The earth magnetic field at a geomagnetic poles has a magnitude $6.2 \times 10^{-5} \text{ T}$. Find magnetic field at this point on the surface where the radius makes an angle of 135° with this axis of the earths assumed to be dipole
 - 1) 4.9×10^{-5}
 - 2) 3.9×10^{-5}
 - 3) 2.9×10^{-5}
 - 4) none
10. The intensities of magnetic field at two points on the axes of a bar magnet at distances 0.1 m, 0.2 m are 12.5:1 from the mid point. Calculate the distance between the poles of a magnet.
 - 1) 0.2 m
 - 2) 0.3 m
 - 3) 0.1 m
 - 4) 0.5 m
11. Two short magnets a, b of moments 0.108 Am^2 and 0.192 Am^2 are placed along mutually perpendicular directions meeting at a point p, find the direction of magnetic field at point p. If it lies at a distance 30 cm and 60 cm respectively from the centres of two magnets.
 - 1) 24°
 - 2) 34°
 - 3) 44°
 - 4) 14°

Numerical Value Type Questions

12. Two short magnets each of moment M are placed one over the other at right angles. The resultant magnetic induction at a point 'P', that lies at a distance 'd' from the common centre is $\sqrt{P} \frac{\mu_0}{4\pi} \frac{3M}{d^3}$ then 'p' is



13. A bar magnet of 10 cm long is kept with its N - pole pointing north. A neutral point is formed at a distance of 15 cm from each pole. Given the horizontal component of earth's field is 0.4 Gauss, the pole strength of the magnet is _____ amps-m

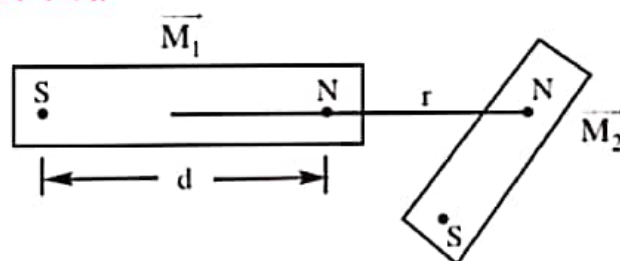
EXERCISE-III

(Couple, p.E and work done)

Straight Objective Type Questions

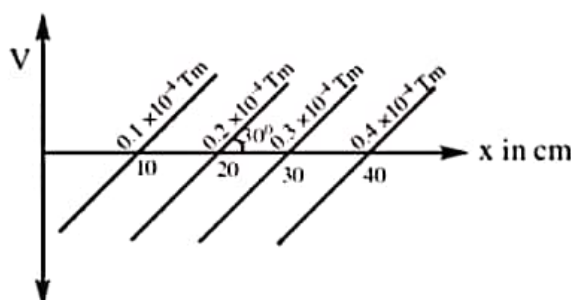
- A magnet is suspended in the magnetic meridian with an untwisted wire. The upper end of the wire is rotated through 180° to deflect the magnet by 30° from magnetic meridian. Now this magnet is replaced by another magnet. Now the upper end of the wire is rotated through 270° to deflect the magnet 30° from the magnetic meridian. The ratio of the magnetic moments of the two magnets is
1) 3 : 4 2) 1 : 2 3) 4 : 7 4) 5 : 8
- A magnetic dipole is under the influence of two magnetic fields. The angle between the field directions is 60° and one of the fields has a magnitude of $1.2 \times 10^{-2} \text{ T}$. If the dipole comes to stable equilibrium at angle of 15° with this field, then the magnitude of the other field ($\sin 15^\circ = 0.2588$)
1) $1.39 \times 10^{-3} \text{ T}$ 2) $2.39 \times 10^{-3} \text{ T}$ 3) $3.39 \times 10^{-3} \text{ T}$ 4) $4.39 \times 10^{-3} \text{ T}$
- If 'W' Joule is the work done in rotating the magnet from the field direction to a position that makes an angle 60° with the direction of the field, the torque required to hold the magnet in that direction is
1) W N-m 2) $\frac{W}{2} \text{ N-m}$ 3) $\frac{W}{\sqrt{3}} \text{ N-m}$ 4) $\sqrt{3} \text{ WN-m}$
- Two small bar magnets having dipole moments M_1, M_2 placed at distance r. Find the magnetic interaction energy of this system of dipoles for $d \ll r$.

- $-\frac{\mu_0 M_1 M_2 \cos \theta}{2\pi r^3}$
- $\frac{\mu_0 M_1 M_2 \sin \theta}{2\pi r^3}$
- $\frac{\mu_0 M_1 M_2 \tan \theta}{2\pi r^3}$
- $\frac{\mu_0 M_1 M_2 \cot \theta}{2\pi r^3}$



- A magnetic dipole with a dipole moment of magnitude 0.020 Am^2 is released from rest in a uniform magnetic field of induction 52 mT. When the dipole rotates due to magnetic couple on it through the orientation where its dipole moment is aligned with the magnetic field its KE is 0.80 mJ. The angle between the dipole moment and magnetic field
1) 46.67° 2) 56.67° 3) 76.67° 4) 76.67°
- A magnetic needle having magnetic moment 10 Am^2 and length 10 cm is clamped at its centre in such a way that it can be rotated in the vertical east and west plane. A horizontal force towards east is applied at the north pole to keep the needle fixed at an angle of 30° with vertical. Find the magnitude of applied force. ($B_V = 40 \mu \text{ T}$)
1) $1.3 \times 10^{-2} \text{ N}$ 2) $4.3 \times 10^{-2} \text{ N}$ 3) $2.3 \times 10^{-2} \text{ N}$ 4) $0.3 \times 10^{-2} \text{ N}$
- Magnetic scalar potential due to a magnetic dipole at a point on its axis situated at a distance of 20 cm from the centre of is found to be $1.2 \times 10^{-5} \text{ Tm}$. Find the magnetic dipole moment of the dipole is
1) 1.8 Am^2 2) 2.8 Am^2 3) 5.8 Am^2 4) 4.8 Am^2

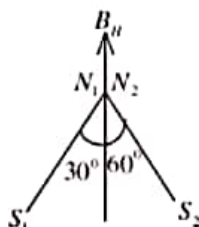
8. Figure shows some equipotential surfaces of the magnetic scalar potential. Find the magnetic field at B at a point in the region.



- 1) 10^{-4} T 2) 3×10^{-4} T 3) 2×10^{-4} T 4) 5×10^{-4} T

Numerical Value Type Questions

9. A short bar magnet of magnetic moment $M = 0.32 \text{ JT}^{-1}$ is placed in a uniform magnetic field of 0.15 T. If the bar is free to rotate in the plane of the field, which orientation would correspond to its (a) stable, and (b) unstable equilibrium? What is the potential energy of the magnet in each case $-4.8 \times 10^{-3} \text{ J}$, $+4.8 \times 10^{-3} \text{ J}$ then x, y are
10. Two small magnets X and Y of dipole moments M_1 and M_2 are fixed perpendicular to each other with their north poles in contact. This arrangement is placed on a floating body so as to move freely in earth's magnetic field as shown in figure then the ratio of magnetic moment is $\sqrt{2^n - 1}$ then n is



11. A bar magnet of magnetic moment 2 Am^2 is free to rotate about a vertical axis passing through its centre. The magnet is released from rest from east - west position. Then the K.E of the magnet as it takes north-south position is $(2n + 40) \mu\text{J}$ then n is ($B_H = 25 \mu\text{T}$)
12. A short bar magnet placed with its axis at 30° with a uniform external magnetic field of 0.16T experiences a torque of magnitude 0.032Nm. If the bar magnet were free to rotate, its P.E when it is in stable and unstable equilibrium are (Only magnitude)

EXERCISE-IV

(Time period of oscillation)

Straight Objective Type Questions

1. A small magnet of moment $4.8 \times 10^{-2} \text{ J/T}$ is suspended freely in the plane of a uniform magnetic field of magnitude $3 \times 10^{-2} \text{ T}$. If the magnet is slightly displaced from its stable equilibrium and released then the angular frequency of its oscillations in rad/sec. (Moment of inertia of the magnet about the axis of rotation is $2.25 \times 10^{-5} \text{ kgm}^2$)
- 1) 8 2) 4 3) 3 4) 2

2. A combination of two identical bar magnets are placed one over the other such that they are perpendicular and bisect each other. The time period of oscillation in a horizontal magnetic field is $2^{5/4}$ seconds. One of the magnets is removed and if the other magnet oscillates in the same field, then the time period in seconds is
 1) $2^{1/4}$ 2) $2^{1/2}$ 3) 2 4) $2^{5/4}$
3. A magnetic needle has a frequency of 20 oscillations per minute in the earth's horizontal field. when the field of a magnet supports the earth's horizontal field, the frequency increases to 30 oscillations per minute. The ratio of the field of the magnet to that of the earth is
 1) 4:7 2) 7:4 3) 5:4 4) 4:5
4. Two long bar magnet of magnetic moment 6Am^2 and 8Am^2 can be fastened together with their axes horizontal and their centres in a vertical line about which the system is free to oscillate. Compare the time periods in the earth's field when the axes of the magnets are (i) in the same direction (ii) in opposite direction (iii) at right angle
 1) 1:7:5 2) $1:\sqrt{7}:\sqrt{5}$ 3) $1:\sqrt{7}:\sqrt{7/5}$ 4) $1:\sqrt{7/5}:\sqrt{14/5}$
5. A compass needle makes 10 oscillations per minute in the earth's horizontal field. A bar magnet deflects the needle by 60° from the magnetic meridian. The frequency of oscillation in the deflected position in oscillations per minute is (field due to magnet is perpendicular to B_H)
 1) $5\sqrt{2}$ 2) $20\sqrt{2}$ 3) $10\sqrt{2}$ 4) 10
6. T_0 is time period of oscillation of the needle in uniform horizontal magnetic field (B_H) of the earth. If another magnetic field is applied perpendicular to B_H such that the needle deflects by θ for its equilibrium. Now period of oscillation of the needle is T. The relation between T and T_0 is :
 1) $T^2 = T_0^2 \cos\theta$ 2) $T = T_0 \cos\theta$ 3) $T = \frac{T_0}{\cos\theta}$ 4) $T^2 = \frac{T_0^2}{\cos\theta}$
7. With a standard rectangular bar magnet the time period of a vibration magnetometer is 4 seconds. The bar magnet is cut parallel to its length into four equal pieces. The time period of vibration magnetometer when one piece is used (in seconds) (bar magnet breadth is small) is
 1) 16 2) 8 3) 4 4) 2
8. A compass needle oscillates 20 times per minute at a place where the dip is 45° and 30 times per minute where the dip is 30° . Compare the total magnetic field due to earth at the two places.
 1) 2.83 2) 1.83 3) 0.83 4) 3.83

Numerical Value Type Questions

9. The magnetic needle of a vibration magnetometer makes 12 oscillations per minute in the horizontal component of earth's magnetic field. When an external short bar magnet is placed at some distance along the axis of the needle in the same line it makes 15 oscillations per minute. If the poles of the bar magnet are inter changed, the number of oscillations it takes per minute is $\sqrt{q \times 9}$ then q is
10. In an uniform field the magnetic needle completes 10 oscillations in 92 seconds. When a small magnet is placed in the magnetic meridian 10cm due north of needle with north pole towards south completes 15 oscillations in 69 seconds. The magnetic moment of magnet ($B_H = 0.3 \text{ G}$) is
11. A short magnet oscillates in vibration magnetometer with a frequency 10Hz where horizontal component of earth's magnetic field is $12\mu\text{T}$. A downward current of 15A is established in the vertical wire placed 20cm west of the magnet. New frequency is $\sqrt{4n+5}$ then n is

EXERCISE-V

(Terrestrial Magnetism)

Straight Objective Type Questions

- The plane of the dip circle is set in the geo-graphical meridian and the apparent dip is θ_1 . It is then set in a vertical plane perpendicular to the geographical meridian, the apparent dip becomes θ_2 . The angle of declination α at that place is given by
 1) $\tan \alpha = \sqrt{\tan \theta_1 \tan \theta_2}$ 2) $\alpha = \sqrt{\tan^2 \theta_1 \tan^2 \theta_2}$ 3) $\tan \alpha = \frac{\tan \theta_1}{\tan \theta_2}$ 4) $\tan \alpha = \frac{\tan \theta_2}{\tan \theta_1}$
- A dip circle lying initially in the magnetic meridian is rotated through angle θ in the horizontal plane. The ratio of tangent of apparent angle of dip to true angle of dip is
 1) $\cos \theta : 1$ 2) $\sin \theta : 1$ 3) $1 : \cos \theta$ 4) $1 : \sin \theta$
- In the magnetic meridian of a certain place, the horizontal component of the earth's magnetic field is 0.26 G and the dip angle is 60° . What is the magnetic field of the earth at this location ?
 1) 0.24 G 2) 0.58G 3) 0.52G 4) 0.33 G
- A dip needle vibrates in the vertical plane perpendicular to the magnetic meridian. The time period of vibration is found to be 2 seconds. The same needle is then allowed to vibrate in the horizontal plane and the time period is again found to be 2 seconds. Then the angle of dip is
 1) 0° 2) 30° 3) 45° 4) 90°
- A ship is to reach a place 10° south of west. In which direction should be steered if the declination at the place is 18° west of north.
 1) 82° 2) 92° 3) 72° 4) none
- True value of dip at a place is 45° . The plane of the dip circle is rotated through 60° from the magnetic meridian. The apparent value of dip
 1) 53.4° 2) 63.4° 3) 43.4° 4) 73.4°

Numerical Value Type Questions

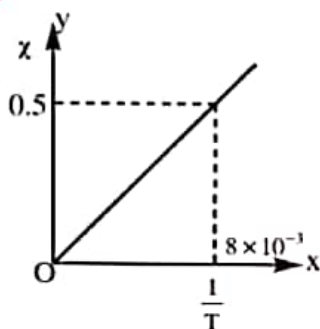
- A magnet makes 12 oscillation per minute at a place where horizontal component of earth's field is $6.4 \times 10^{-3} \text{ T}$. It is found to require 8 seconds per oscillation at another place X. The vertical component of earth's field at X where resultant field makes angle 60° with horizontal is $\frac{25}{x} \times 10^{-4} \text{ T}$ then x is
- Obtain the earth's magnetisation, assuming that the earth's field can be approximated by a giant bar magnet of magnetic moment $8.0 \times 10^{22} \text{ Am}^2$. Radius of earth = 6400km.
- The horizontal component of earth's magnetic field at place is $0.36 \times 10^{-4} \text{ weber/m}^2$. If the angle of dip at that place is 60° then the value of vertical component of earth's magnetic field will be $1.2\sqrt{x}\sqrt{3} \times 10^{-5} \text{ T}$ then x is (in wb/m^2)

EXERCISE-VI

(Properties of magnetic materials, Hysterin loop, χ , Curie Law)Straight Objective Type Questions

- A rod of ferromagnetic material with dimensions $10\text{cm} \times 0.5\text{cm} \times 2\text{cm}$ is placed in a magnetising field of intensity $2 \times 10^5 \text{ A/m}$. The magnetic moment produced due it is 6 amp-m^2 . The value of magnetic induction will be ----- 10^{-2} T .
 1) 100.48 2) 200.28 3) 50.24 4) 300.48

2. A magnetic material of volume 30cm^3 is placed in a magnetic field of intensity 5 oersted. The magnetic moment produced due it is 6 amp-m^2 . The value of magnetic induction will be.
 1) 0.2517 Tesla 2) 0.025 Tesla 3) 0.0025 Tesla 4) 25 Tesla
3. The total magnetic flux in a material, which produces a pole of strength m_p when a magnetic material of cross-sectional area A is placed in a magnetic field of strength H , will be
 1) $\mu_0(AH+m_p)$ 2) μ_0AH 3) μ_0m_p 4) $\mu_0(m_pAH+A)$
4. A magnetizing field of 1600 Am^{-1} produces a magnetic flux of 2.4×10^{-5} weber in a bar of iron of cross - section $0.2 \times 10^{-4}\text{m}^2$. Calculate permeability and susceptibility of the bar.
 1) $7.5 \times 10^{-5}\text{ TA}^{-1}\text{m}$, 496.1 2) $75 \times 10^{-5}\text{ TA}^{-1}\text{m}$, 596.1
 3) $1.5 \times 10^{-4}\text{ TA}^{-1}\text{m}$, 596.1 4) $1.5 \times 10^{-4}\text{ TA}^{-1}\text{m}$, 396.1
5. The core of a toroid having 3000 turns has inner and outer radii of 11cm and 12cm respectively. The magnetic flux density in the core for a current of 0.70A is 2.5T. What is the relative permeability of the core?
 1) 484 2) 584 3) 384 4) 684
6. A magnetising field of 1500 A/m produces a magnetic flux of $2.4 \times 10^{-5}\text{ wb}$ in a bar of iron cross section 0.5 cm^2 . Calculate the permeability.
 1) $3.2 \times 10^{-4}\text{ Tm/A}$ 2) $2.2 \times 10^{-4}\text{ Tm/A}$ 3) $4.2 \times 10^{-4}\text{ Tm/A}$ 4) $6.2 \times 10^{-4}\text{ Tm/A}$
7. The coercivity of certain permanent magnet is $4 \times 10^4\text{ A/m}$. This magnet is placed in side a solenoid 15 cm long and having 600 turns and a current is passed in the solenoid to demagnetise it completely find the current.
 1) 20A 2) 30A 3) 10A 4) 40A
8. The susceptibility of magnesium at 300 K is 1.2×10^{-5} . At what temperature will be the susceptibility increases to 1.8×10^{-5} .
 1) 300K 2) 200K 3) 400K 4) 500K
9. Find curie constant from it χ vs $\frac{1}{T}$ graph shown in the figure.



- 1) 62.5 K 2) 72.5 K 3) 82.5 K 4) 42.5K

Numerical Value Type Questions

10. An iron rod is subjected to cycles of magnetisation at the rate of 50Hz. Given the density of the rod is $8 \times 10^3\text{kg/m}^3$ and specific heat is $0.11 \times 10^3\text{ cal/kg}^\circ\text{C}$. The rise in temperature per minute, if the area enclosed by the B – H loop corresponds to energy of 10^{-2} J is (Assume there is no radiation losses) $(\sqrt{8^y} + 0.1)^\circ\text{C}$ then y is
11. A magnetic dipole is placed under the effect of two magnetic fields inclined at 75° to each other. One of the fields has a magnetic induction of $1.5 \times 10^{-2}\text{ T}$. The magnet comes to rest at an angle of 30° with the direction of this field. The magnetude of the other field is $\frac{3}{\sqrt{n}} \times 10^{-2}\text{ T}$ then n is

KEY SHEET (LECTURE SHEET)

EXERCISE-I

1) 2 2) 4 3) 2 4) 2

EXERCISE-II

1) 3 2) 4 3) 1 4) 2 5) 4 6) 2 7) 3 8) 2
9) 1 10) 3 11) 4 12) 3 13) 13.5

EXERCISE-III

1) 4 2) 4 3) 4 4) 1 5) 3 6) 3 7) 4 8) 3
9) $x=2, y=2$ 10) 2 11) 5 12) +0.064, 0.064

EXERCISE-IV

1) 1 2) 3 3) 3 4) 3 5) 3 6) 1 7) 3 8) 2
9) 7 10) 0.75 11) 5

EXERCISE-V

1) 3 2) 3 3) 3 4) 3 5) 1 6) 2 7) 3
8) 72.8 Am^{-1} 9) 9

EXERCISE-VI

1) 1 2) 1 3) 1 4) 2 5) 4 6) 1 7) 3 8) 2
9) 1 10) 2 11) 8